Logical Fallacy Detection

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arXiv:2202.13758v3 [cs.CL] 12 Dec 2022

Abstract

Reasoning is central to human intelligence. However, fallacious arguments are common, and some exacerbate problems such as spreading misinformation about climate change. In this paper, we propose the task of *logical fallacy detection*, and provide a new dataset (**LOGIC**) of logical fallacies generally found in text, together with an additional challenge set for detecting logical fallacies in climate change claims (**LOGICCLIMATE**). Detecting logical fallacies is a hard problem as the model must understand the underlying logical structure of the argument. We find that existing pretrained large language models perform poorly on this task. In contrast, we show that a simple structure-aware classifier outperforms the best language model by 5.46% F_1 scores on LOGIC and 4.51% on LOGICCLIMATE. We encourage future work to explore this task since (a) it can serve as a new reasoning challenge for language models, and (b) it can have potential applications in tackling the spread of misinformation.[1]

1 Introduction

Reasoning is the process of using existing knowledge to make inferences, create explanations, and generally assess things rationally by using logic (Aristotle, 1991). Human reasoning is, however, often marred with logical fallacies. Fallacious reasoning leads to disagreements, conflicts, endless debates, and a lack of consensus. In daily life, fallacious arguments can be as harmless as "*All tall people like cheese*" (faulty generalization) or "*She is the best because she is better than anyone else*" (circular claim). However, logical fallacies are also intentionally used to spread misinformation, for instance "*Today is so cold, so I don't believe in global warming*" (faulty generalization) or "*Global warming doesn't exist because the earth is not getting warmer*" (circular claim).

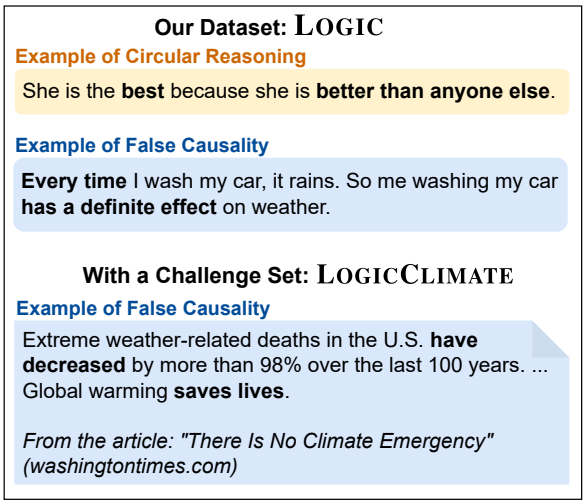

Figure 1: Our dataset consists of general logical fallacies (LOGIC) and an additional test set of logical fallacies in climate claims (LOGICCLIMATE).

In order to detect such fallacious arguments, we propose the task of *logical fallacy detection*. Logical fallacy detection methods can be helpful to tackle important social problems. For instance, these methods can be combined with fact-checkers (Riedel et al., 2017; Thorne et al., 2018) for misinformation detection as many claims can be factually correct but still fallacious. However, logical fallacy detection is challenging as it requires a model to discover egregious patterns of reasoning (Johnson and Blair, 2006; Damer, 2009).

To address this pressing need and encourage more work to detect reasoning flaws, we construct a dataset of logical fallacies consisting of general logical fallacies (**LOGIC**), and a challenging extrapolation set of climate claims (**LOGICCLIMATE**), as shown in Figure 1. We find that this task is challenging for 12 pretrained large language models, whose performances range from 8.62% to 53.31% micro F_1 scores on the LOGIC dataset.

[*] Equal contribution.
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[1]Our dataset and code are available at https://github.com/causalNLP/logical-fallacy.

Logical Fallacy	Examples
Faulty Generalization (18.01%)	"I met **a tall man** who loved to eat cheese. Now I believe that **all tall people** like cheese." "**Sometimes** flu vaccines don't work; therefore **vaccines are useless**."
Ad Hominem (12.33%)	"What can our new math teacher know? Have you seen **how fat** she is?" "I cannot listen to anyone **who does not share my social and political values**."
Ad Populum (9.47%)	"Everyone should like coffee: **95%** of teachers do!" "Killing thousands of people as a result of drug war campaign is **not a crime** to humanity because **millions of Filipino support it**."
False Causality (8.82%)	"**Every time** I wash my car, it rains. Me washing my car has **a definite effect** on the weather." "Every severe recession **follows** a Republican Presidency; therefore Republicans **are the cause of** recessions."
Circular Claim (6.98%)	"J.K. Rowling is a **wonderful** writer because she **writes so well**." "She is the **best** candidate for president because she **is better than** the other candidates!"
Appeal to Emotion (6.82%)	"It is an **outrage** that the school wants to remove the vending machines. This is **taking our freedom away**!" "Vaccines are so **unnatural**; it's **disgusting** that people are willing to put something like that in their body."
Fallacy of Relevance (6.61%)	"**Why are you worried** about poverty? **Look how many** children we abort every day." "**Why should we be worrying** about how the government treats Native people, **when** people in our city can't get a job"
Deductive Fallacy (6.21%)	"**It is possible** to fake the moon landing through special effects. **Therefore**, the moon landing was a fake using special effects." "Guns **are like** hammers—they're both tools with metal parts that could be used to kill someone. And **yet it would be ridiculous** to restrict the purchase of hammers, **so** restrictions on purchasing guns are **equally ridiculous**."
Intentional Fallacy (5.84%)	"**No one has** ever been able to prove that extraterrestrials exist, **so they must** not be real." "**It's common sense** that if you smack your children, they will stop the bad behavior. **So don't tell me** not to hit my kids."
Fallacy of Extension (5.76%)	"Their support of the discussion of sexual orientation issues is dangerous: **they advocate for** the exposure of children to sexually explicit materials, **which is wrong**." "They say we should cut back the defense budget. **Their position is that** they want to leave our nation completely defenseless!"
False Dilemma (5.76%)	"You're **either** for the war **or** against the troops." " **I don't want to** give up my car, so **I don't think** I can support fighting climate change."
Fallacy of Credibility (5.39%)	"My **professor**, who has **a Ph.D. in Astronomy**, once **told me** that ghosts are real. **Therefore, ghosts are real.**" "My **minister** says the Covid vaccine will cause genetic mutations. He **has a college degree**, and is a **holy man**, **so he must be right**."
Equivocation (2.00%)	"I don't see how you can say you're an **ethical person**. It's so hard to get you to do anything; your **work ethic** is so bad" "It is **immoral to kill an innocent human** being. Fetuses are innocent human beings. Therefore, it is **immoral to kill fetuses**."

Table 1: Examples of the 13 logical fallacy types and their distribution in the LOGIC dataset. To illustrate of the potential impact of learning logical fallacies, we select some examples with **neutral** impact that we manually identify, and some with **potentially negative** impact.

By analyzing our collected dataset, we identify that logical fallacies often rely on certain false patterns of reasoning. For example, a typical pattern in *false causality* in Figure 1 is "α co-occurs with $\beta \Rightarrow \alpha$ causes β." Motivated by this, we develop an approach to encourage language models to identify these underlying patterns behind the fallacies. In particular, we design a **structure-aware model** which identifies text spans that are semantically similar to each other, masks them out, and then feeds the masked text instances to a classifier. This structure distillation process can be implemented atop any pretrained language model. Experiments show that our model outperforms the best pretrained language model by 5.46% on LOGIC, and 4.51% on LOGICCLIMATE.

In summary, this paper makes the following contributions:

1. We propose a new task of logical fallacy classification.
2. We collect a dataset of 2,449 samples of 13 logical fallacy types, with an additional challenge set of 1,109 climate change claims with logical fallacies.
3. We conduct extensive experiments using 12 existing language models and show that these models have very limited performance on detecting logical fallacies.
4. We design a structure-aware classifier as a baseline model for this task, which outperforms the best language model.

5. We encourage future work to explore this task and enable NLP models to discover erroneous patterns of reasoning.

2 Logical Fallacy Dataset

First, we introduce our data. Our logical fallacy dataset consists of two parts: a) a set of common logical fallacies (LOGIC), and b) an additional challenge set of logically fallacious claims about climate change (LOGICCLIMATE).

2.1 Common Logical Fallacies: LOGIC

Data Collection The LOGIC dataset consists of common logical fallacy examples collected from various online educational materials meant to teach or test the understanding of logical fallacies among students. We automatically crawled examples of logical fallacies from three student quiz websites, Quizziz, study.com and ProProfs (resulting in around 1.7K samples), and manually collected fallacy examples from some additional websites recommended by Google search (resulting in around 600 samples). More data collection and filtering details are in Appendix A.2.

	# Samples	# Sents	# Tokens	Vocab
Total Data	2,449	4,934	71,060	7,624
Train	1,849	3,687	53,475	6,634
Dev	300	638	8,690	2,128
Test	300	609	8,895	2,184

Table 2: Statistics of the LOGIC dataset.

The entire LOGIC dataset contains 2,449 logical fallacy instances across 13 logical fallacy types. We randomly split the data into train, dev, and test sets; dataset statistics are shown in Table 2, and the distribution and examples of each type in Table 1. More details of each fallacy type are in Appendix A.3.

Comparison with Existing Datasets. Due to the challenges of data collection, all previous existing datasets on argument quality are of limited size. In Table 3, we draw a comparison among our dataset and two existing datasets: an argument sufficiency classification dataset (Stab and Gurevych, 2017), which proposes a binary classification task to identify whether the evidence can sufficiently support an argument, and another dataset dedicated for a specific type of logical fallacy called *ad hominem*, or *name-calling* (Habernal et al., 2018b) where the arguer attacks the person instead of the claim.

Dataset	# Claims	# Classes	Purpose
Arg. Suff.	1,029	Binary	Detect insufficiency
Ad Homi.	2,085	Binary	Detect name calling
LOGIC	**2,449**	**Multiple**	Detect **all** fallacy types

Table 3: Comparison of our logical fallacy dataset with two existing datasets, argument sufficiency classification (Stab and Gurevych, 2017) and ad hominem classification (Habernal et al., 2018b).

Compared to the existing datasets, our dataset has two advantages: (1) we have a larger number of claims in our dataset, and (2) our task serves the more general purpose of detecting all fallacy types instead of a single fallacy type. These two characteristics make our dataset significantly more challenging.

2.2 Challenge Set: LOGICCLIMATE

Logical fallacy detection on climate change is a small step towards promoting consensus and joint efforts to fight climate change. We are interested in whether models learned on the LOGIC dataset can generalize well to real-world discussions on climate change. Hence, we collect an extrapolation set LOGICCLIMATE which consists of all climate change news articles from the Climate Feedback website[2] by October 2021.

For each news article, we ask two different annotators who are native English speakers to go through each sentence in the article, and label all logical fallacies if applicable. Since directly classifying the logical fallacies at the article level is too challenging, we let the annotators select the text span while labeling the logical fallacies, and we compose each sample using the sentence containing the selected text span as logical fallacies. Details of the annotation process are described in Appendix A.4.

In total, the LOGICCLIMATE dataset has 1,079 samples of logical fallacies with on average 35.98 tokens per sample, and a vocabulary of 5.8K words. The label distributions are in Table 4. We provide examples of each fallacy in LOGICCLIMATE in Appendix A.5.

3 A Structure-Aware Model

The task of logical fallacy classification is unique in that logical fallacies are not just about the content words (such as the sentiment-carrying words in a

[2] https://climatefeedback.org/feedbacks/